



PROJ703

International

Robotics Challenge

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The Challenge: Eurobot 2026



Winter is coming, and it's time for the squirrels to stock up on hazelnuts so they can spend the whole winter in their soft, warm nests.

But evil humans already took them all, put them in crates and are about to leave with them, and it will let squirrels starving for the next months!

Luckily, it's coffee time for the humans, and the squirrels have only few seconds to grab the hazelnut crates and hide it from the evil humans.

But will they have enough to survive the winter?

cf. Eurobot Rules Beta v0.9

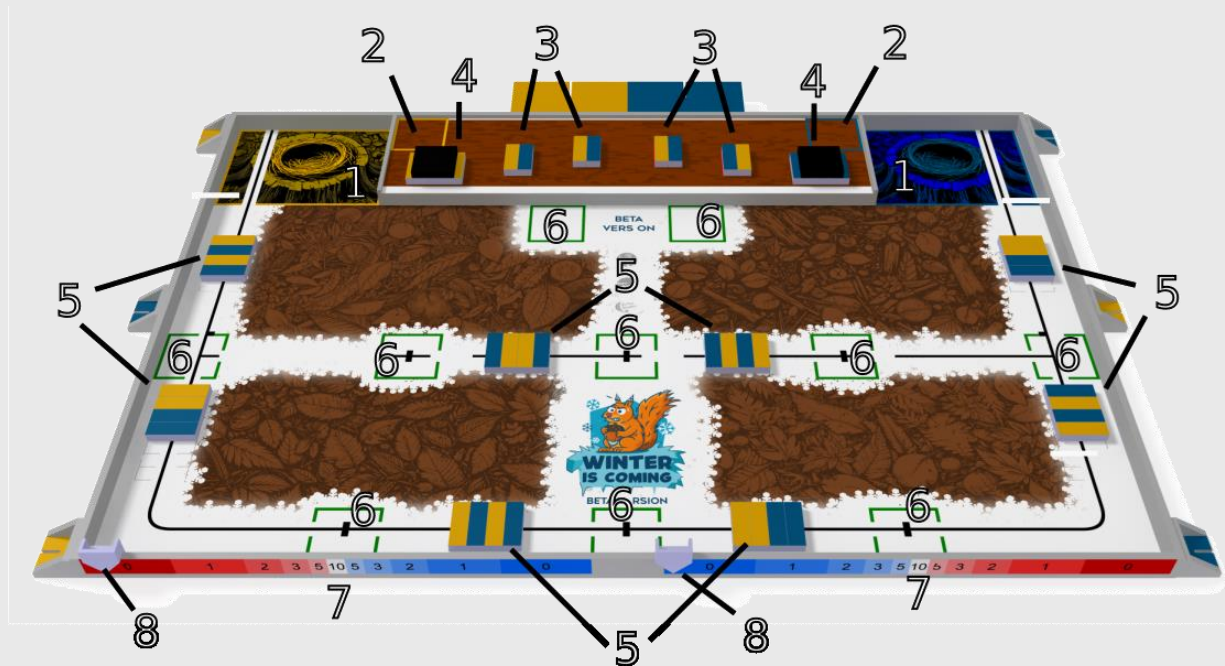
The Strategies



- ***Let's keep the hazelnuts warm***
Collect as many hazelnut crates as possible.
Deposit them in a pantry or in the squirrel nest.
- ***To find is to keep!***
Release one SIMA in the granary.
Have it empty the fridges and take the crates out of the granary.
Fill the fridges with the rotten hazelnuts.
- ***Not too warm, not too cold***
Place the temperature cursor as close to the centre of the thermometer as possible.
- ***Nest, sweet nest***
At the end of the match, the robots must be stopped in their nest.
- ***Lunch time!***
Squirrels must eat their hazelnut reserves to survive the winter.

cf. [Eurobot Rules Beta v0.9](#)

The Playing Area



- Surface: 3000 × 2000 mm²
 - Height: 70 mm
 - Outer side borders: 22 mm
1. Squirrels nest
 2. SIMA Starting areas in the granary
 3. Fridge
 4. Loading area
 5. Collection area
 6. Pantry
 7. Thermometer
 8. Cursor

cf. [Eurobot Rules Beta v0.9](#)

The Challenge

Each team must design a unique and original set of robots, specific to their team. This set includes:

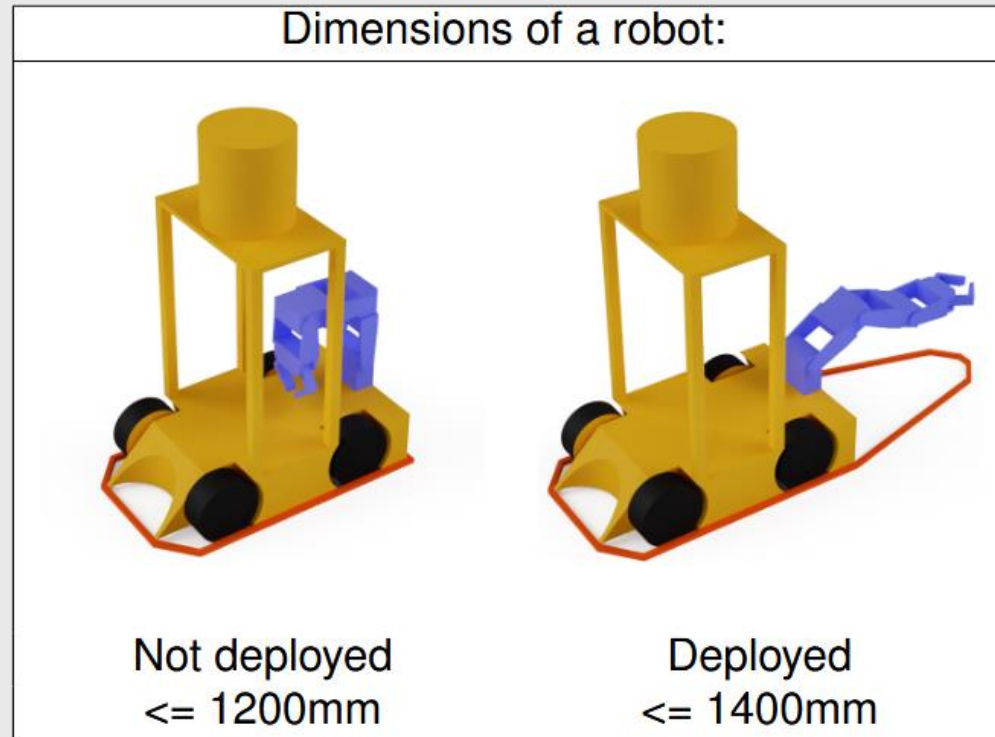
- An autonomous main robot
- One or more autonomous SIMA (optional)
- A remote computing and observation device (optional)
- An embedded/on-board beacon (optional)
- Fixed beacons (optional)

Everything should be ready before
⚠ May 2026 ⚠

cf. Eurobot Contest General Rules v1.2



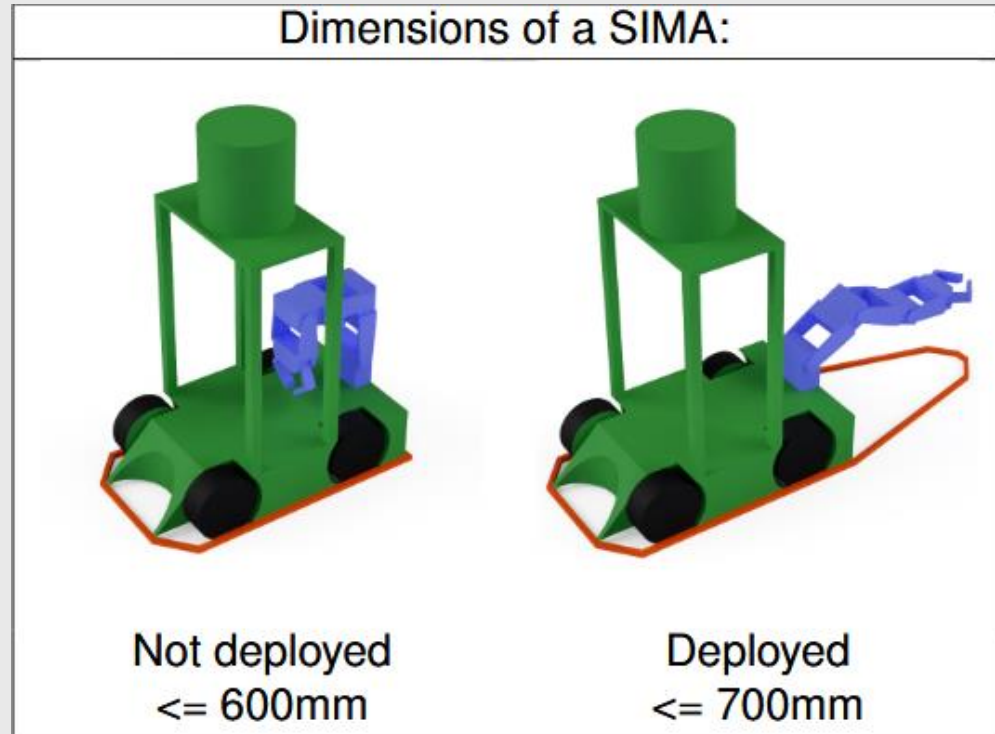
The Main Robot



- Max. perimeter (undeployed): 1200 mm
- Max. perimeter (deployed): 1400 mm
- Max. height: 350 mm
Max. height + E-stop button: 375 mm
- All systems (robots, beacons and accessories) must comply with all applicable regulations in Europe and in the countries hosting the events.

cf. [Eurobot Contest General Rules v1.2](#)

The SIMA (Small Independent Mobile Actuator)



SIMA must respect the following constraints:

- Initial height limit: 150 mm
- Min. size: 60 mm side cube
- Max. perimeter (undeployed): 600 mm
- Max. perimeter (deployed): 700 mm
- Deploy and move height limit: 350 mm
- Weight limit: 1.5 kg
- Cannot be activated by an external element outside of the table (team crew, remote control, ...)

cf. [Eurobot Contest General Rules v1.2](#)

The Beacons

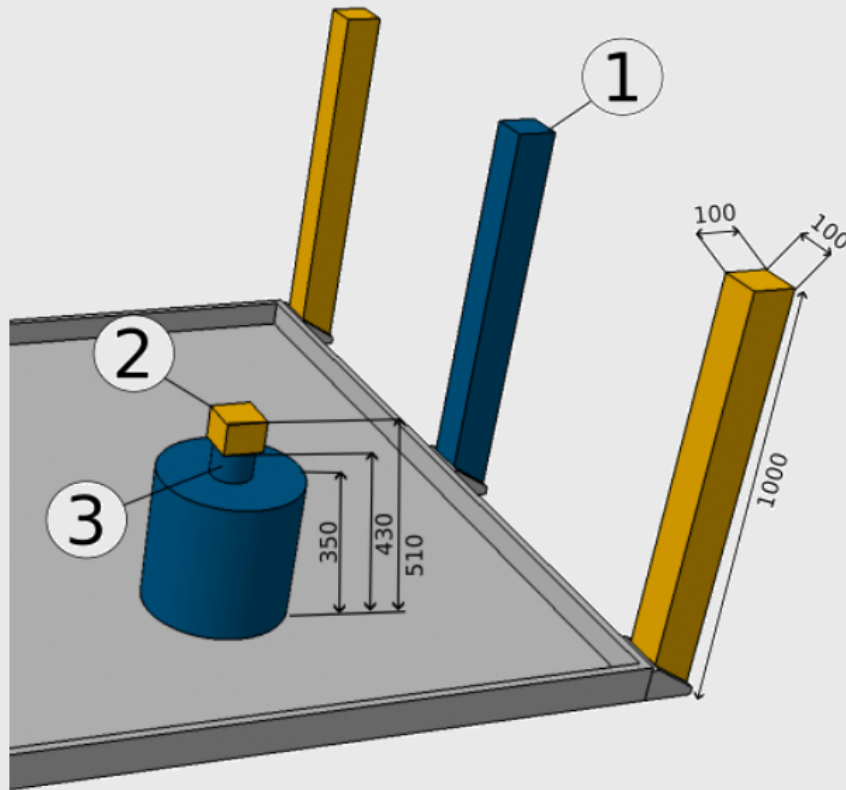


Figure 11 - Example of beacons positioning

Fixed beacons

- Max. dimensions: $100 \times 100 \times 1000 \text{ mm}^3$
- Max. weight: 1.5 kg

Embedded beacons

- Placed on a dedicated mast
- Height: $430 \pm 5 \text{ mm}$ above the playing area
- Hook-and-loop fix

cf. [Eurobot Contest General Rules v1.2](#)

Remote Computing and Observation Device

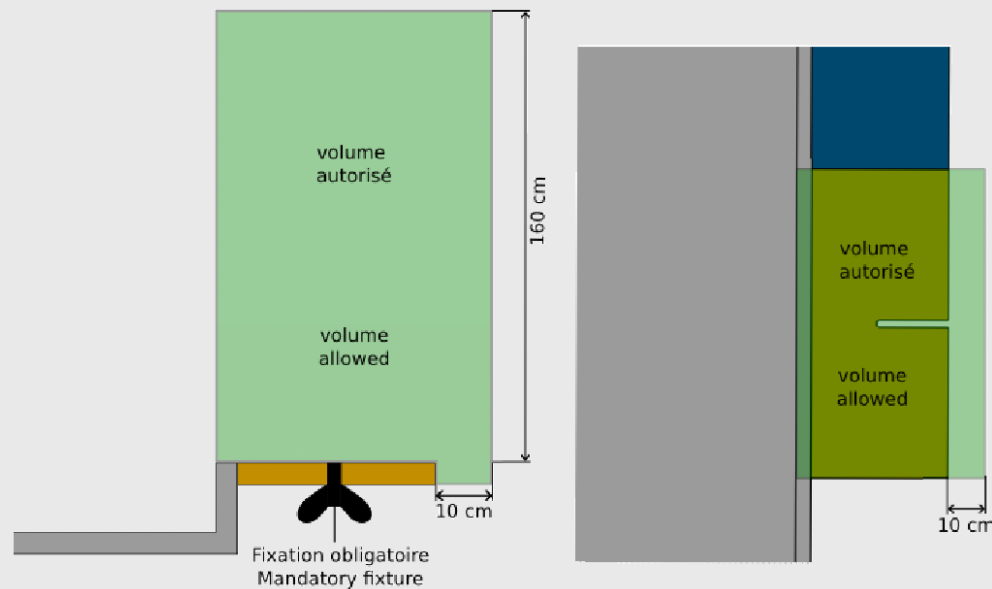


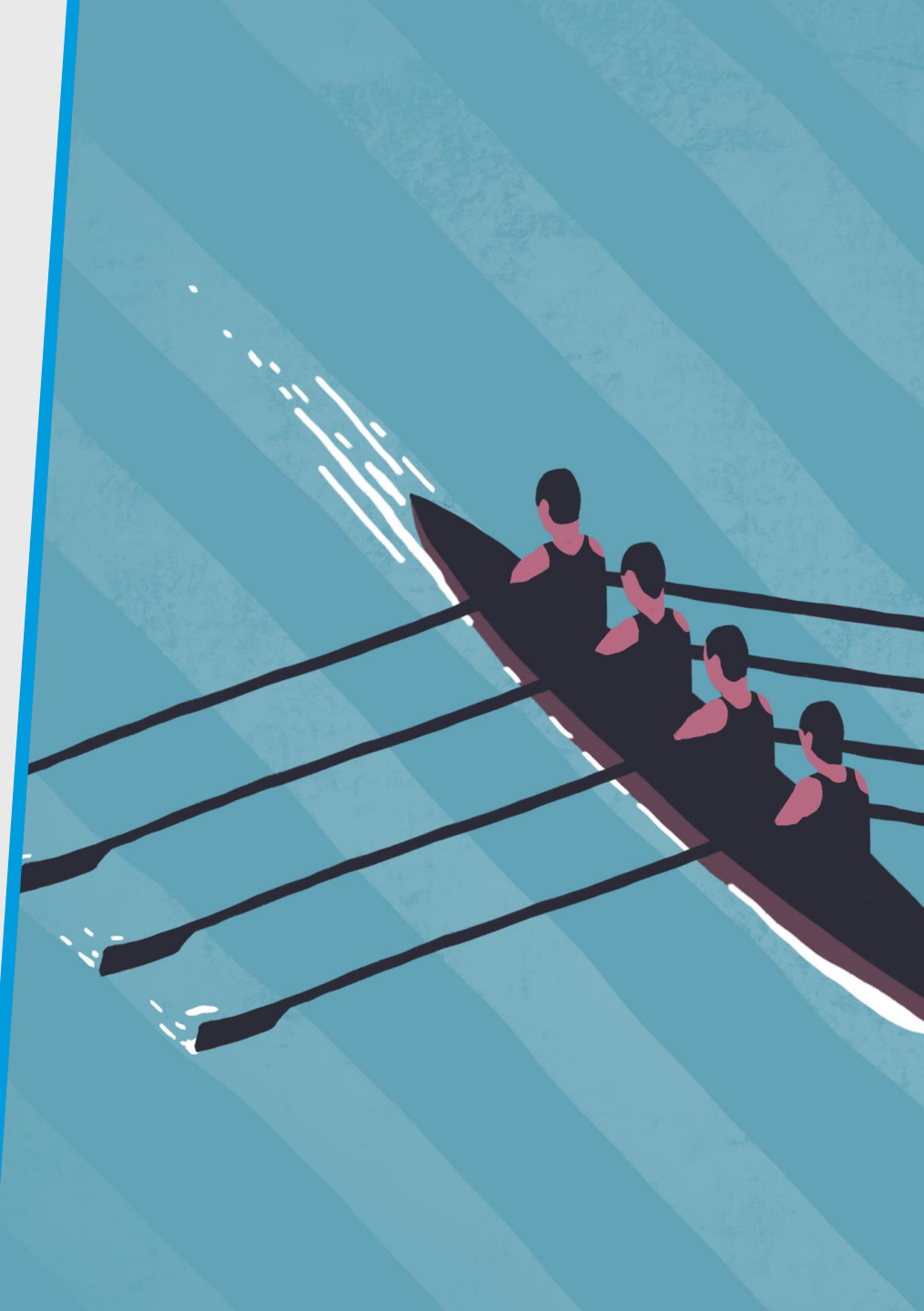
Figure 13 - How to install a remote computing device

- Must be placed on the part of the platform in the colour of the team.
- Must not rise beyond a horizontal plane situated 1.6 m above the upper surface of the platform, and must not descend below the lower surface of the platform of 22 mm, i.e., an area of $450 \times 320 \text{ mm}^2$.
- On the sides, only an offset of 100 mm at the rear of the platform is authorized. Its mass must be less than 5 kg.

cf. Eurobot Contest General Rules v1.2



All hands on deck!





MathWorks Minidrone Competition Europe 2025

No participation fees
Free self-paced tutorials
Complimentary MathWorks software

Steps to compete

1. **Apply** to receive all the competition materials.
2. **Watch** the *MathWorks Minidrone Competition* video series to better understand the competition details.
3. **Review** the competition rules and guidelines.
4. **Complete** *MATLAB Onramp*, *Simulink Onramp*, and *Stateflow Onramp* courses before starting to work on your algorithm.

Competition Timeline

Task	Deadline
Application	01-Oct-2025
Simulation Submission	05-Oct-2025
Finalists Declaration	15-Oct-2025
Final Round	06 & 07 Nov 2025



Finals will be held at the MathWorks Cambridge Office, UK